



Artificial Intelligence

Project 2

Industrial Training Kit

Batch: 2026 | Powered by DecodeLabs

WELCOME TO THE TEAM! 🚀

Step into the role of an Artificial Intelligence Engineer at DecodeLabs. Project 2 is your predictive phase: Data Classification Using AI. This track isn't about "simple rules" anymore—it's about Supervised Learning. Before you dive into complex neural networks, you must master the fundamental pipeline of teaching a machine to recognize patterns in data and categorize new information based on what it has learned. By completing this milestone, you are proving you can train, test, and validate an AI model through pure algorithmic logic. Let's design a digital foundation that empowers machines to make intelligent decisions with absolute precision.



GUIDELINES:

Qualification Criteria

EARN YOUR BADGE 

- Requirement: Complete project 2 to unlock other projects for next week.
- Standard: All projects must be verified for quality.
- Mandatory: Project 2 is the vital bridge to professional development for every intern.



Project 2: Data Classification Using AI

Goal:

Build a basic classification model
using a small dataset.

Key Requirements:

- Load and understand a dataset
 - Split data into training and testing sets
- Apply a simple classification algorithm

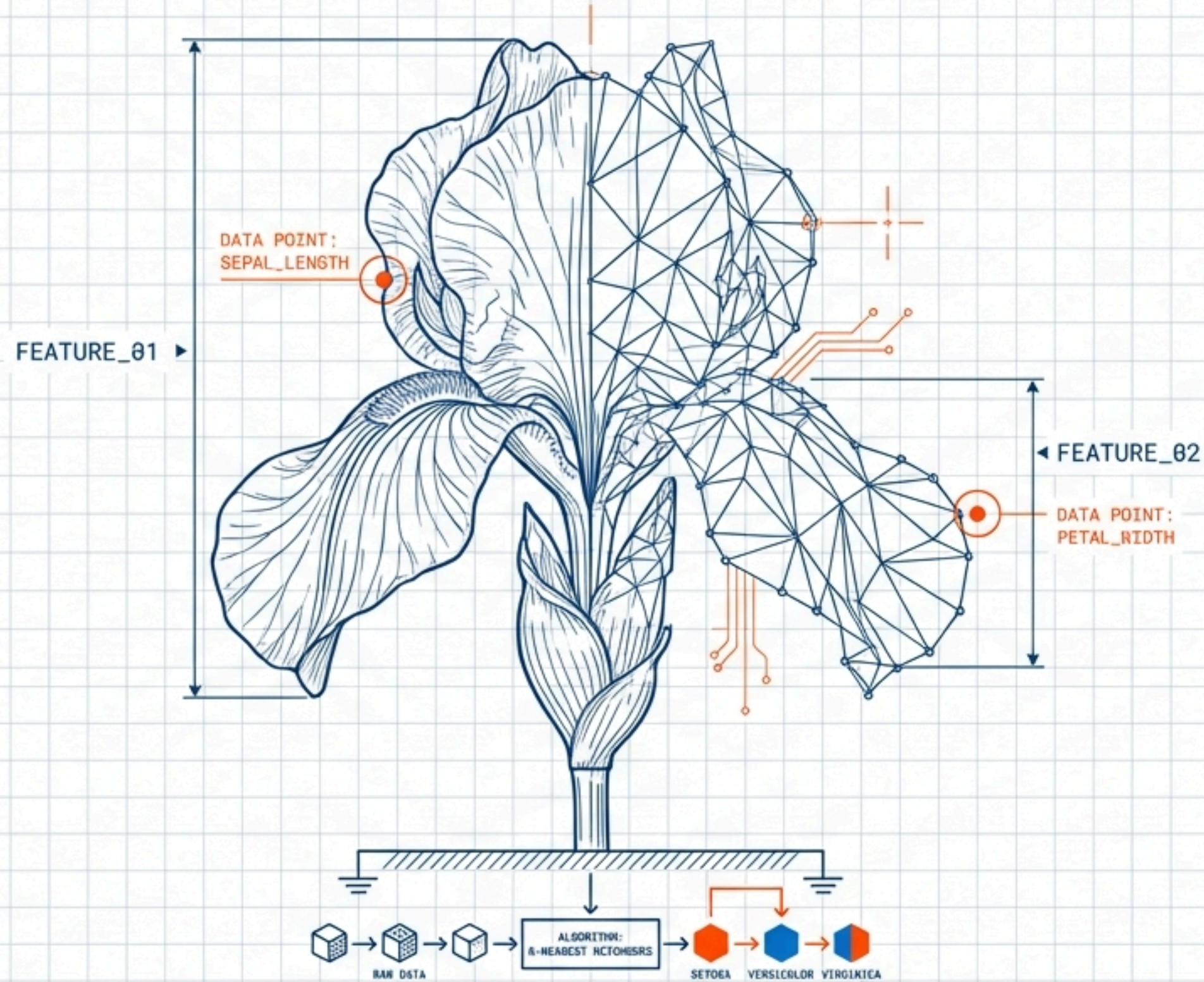
Key Skills:

Data handling, supervised
learning basics, model training



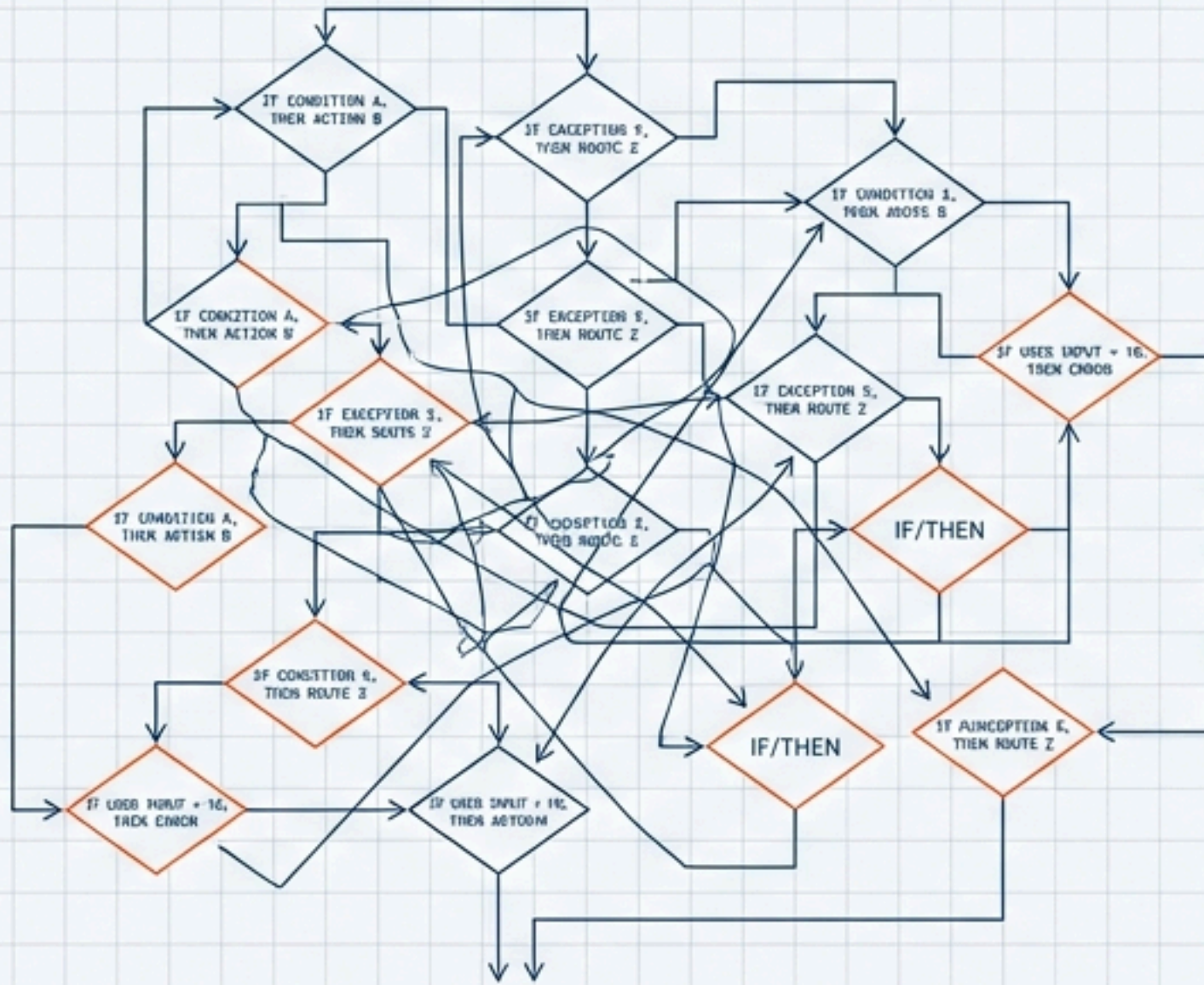
ARCHITECTURAL PARADIGMS

PROJECT 2: DATA CLASSIFICATION // FROM RAW DATA TO INTELLIGENT DECISION MAKING



THE LOGIC SKELETON

OLD WAY: HEURISTIC



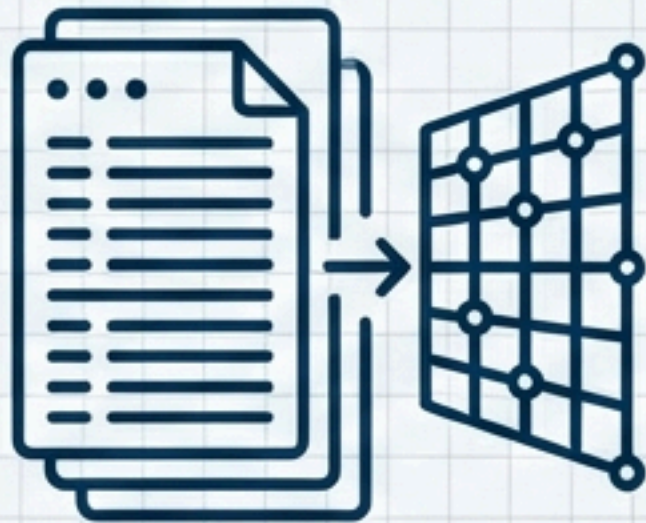
NEW WAY: SUPERVISED LEARNING



We do not write the rules. We provide history, and the machine derives the logic.

THE MASTER BLUEPRINT: IPO FRAMEWORK

INPUT



- Iris Domain
- Feature Scaling

PROCESS



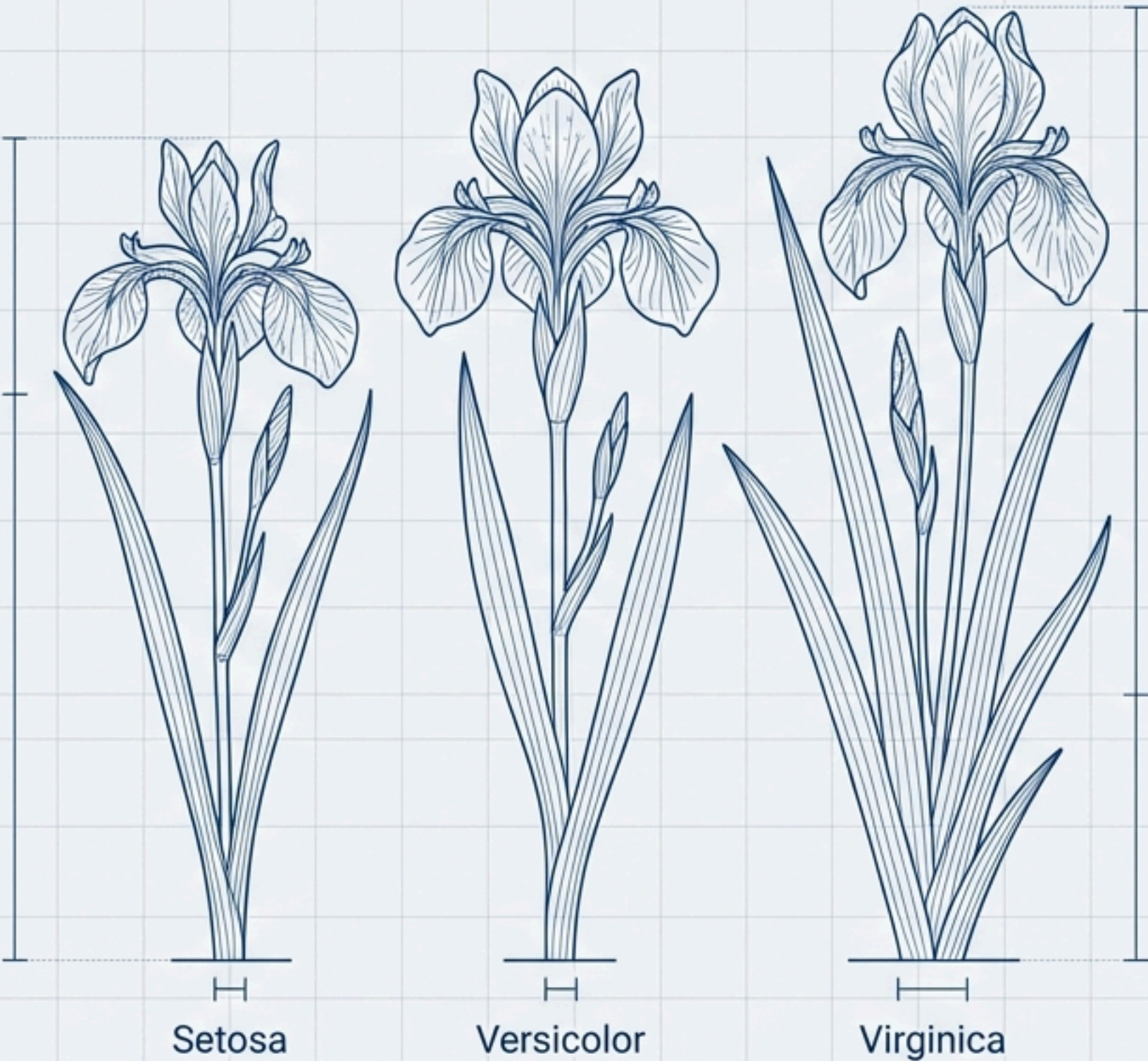
- Train-Test Split
- KNN Algorithm

OUTPUT

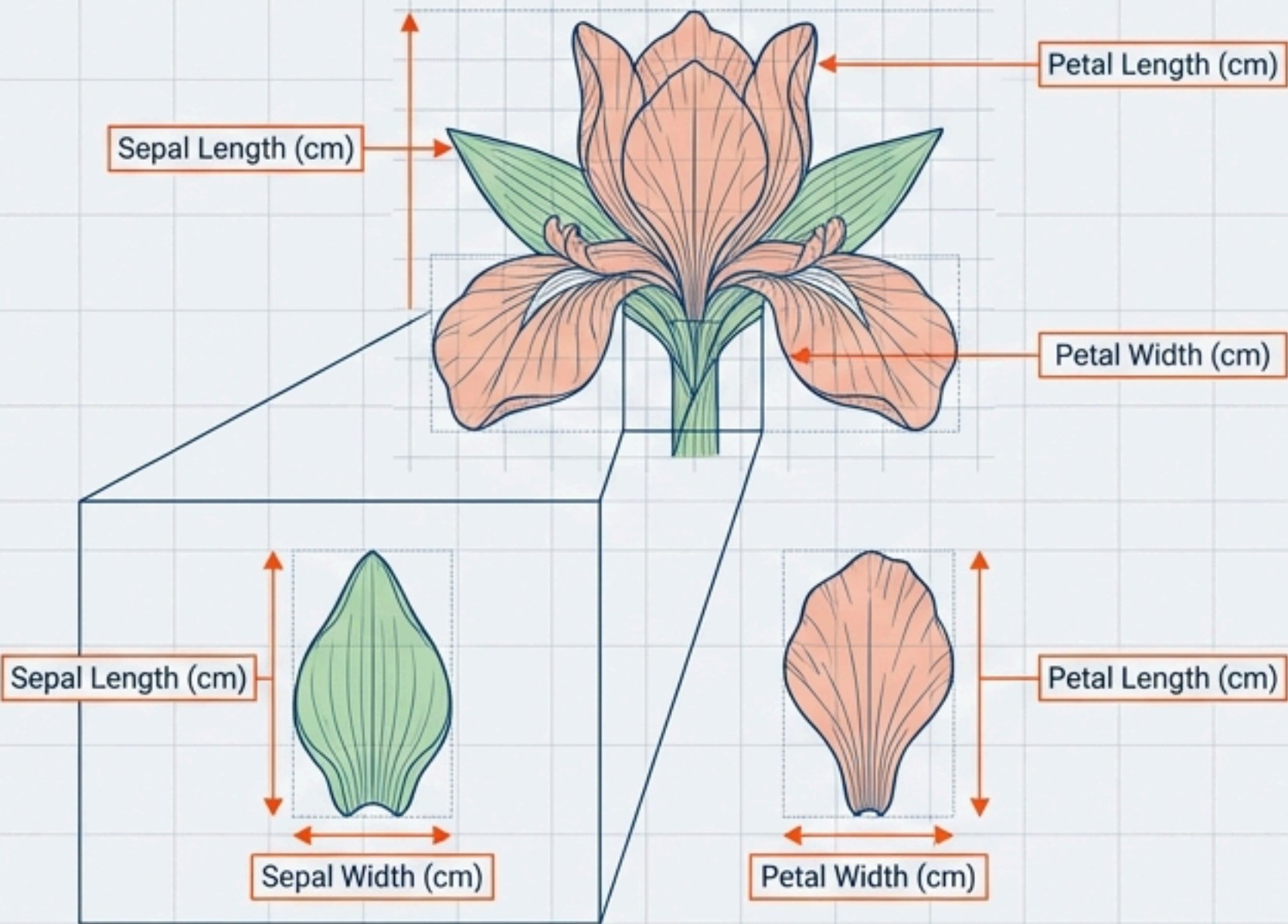


- Confusion Matrix
- F1 Score

RAW MATERIAL: THE IRIS BENCHMARK

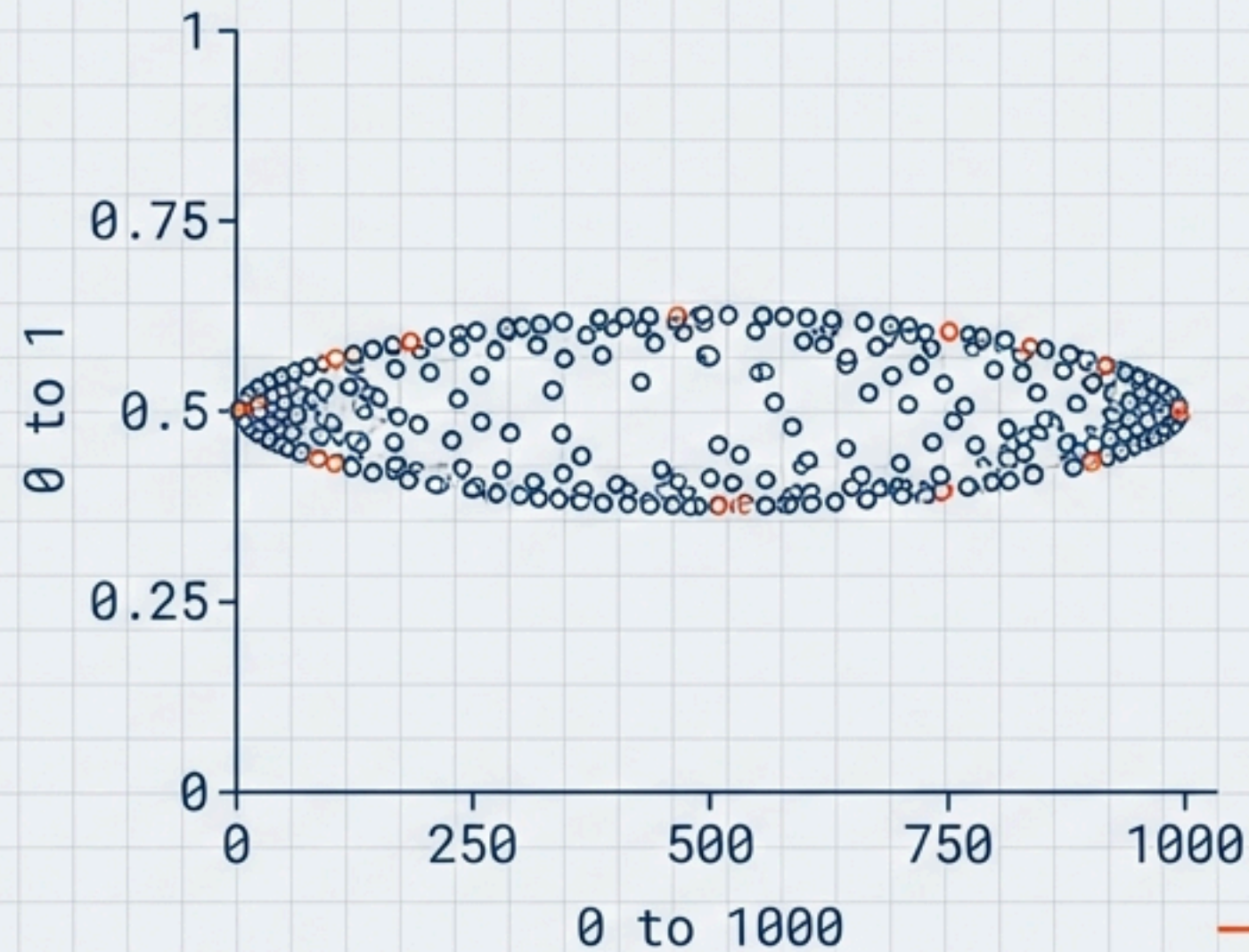


SAMPLES:	150 (Balanced)
CLASSES:	3
DIMENSIONS:	4

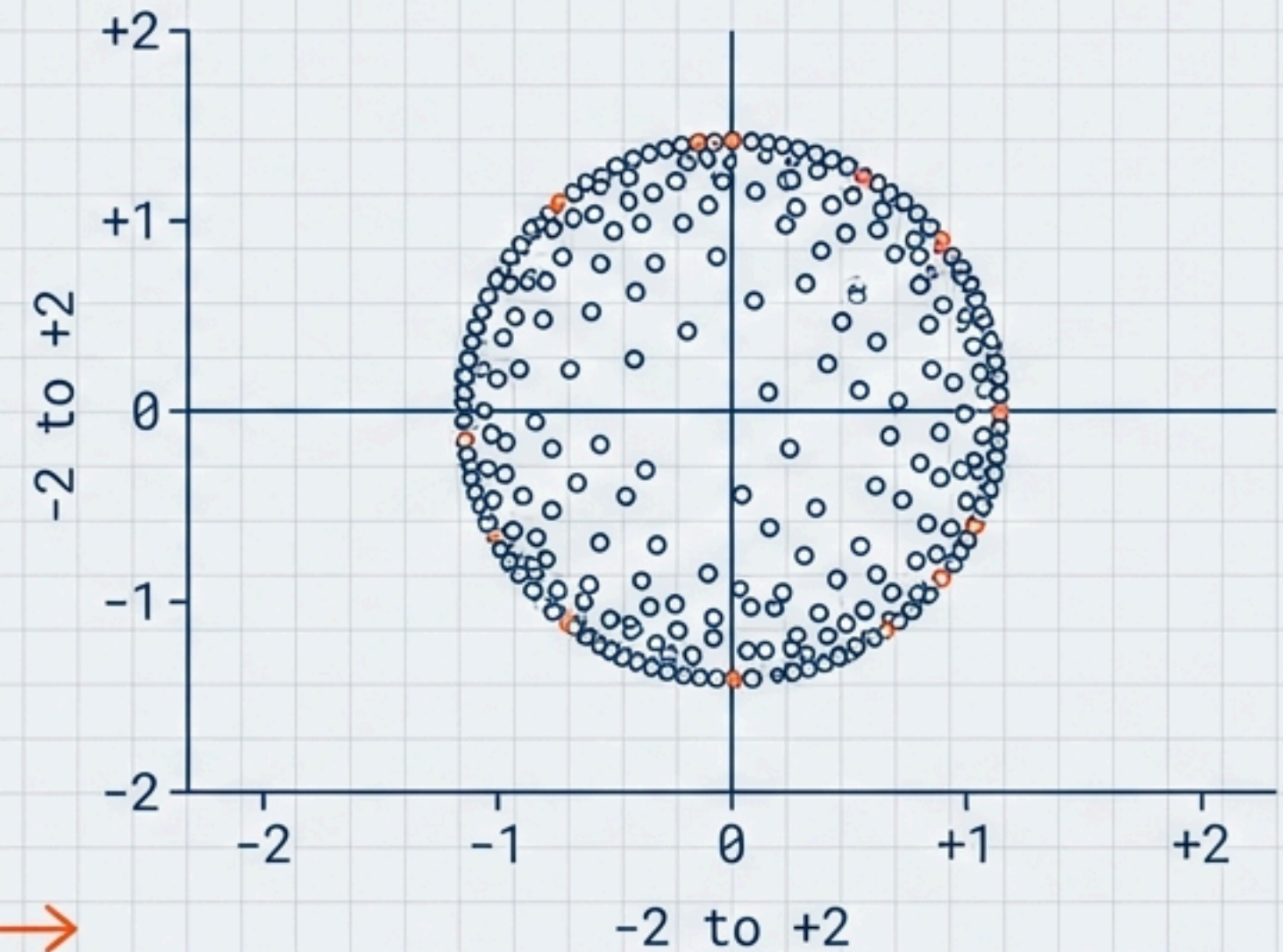


THE GATEKEEPER RULE: SCALING

RAW DATA (BIASED)



STANDARD SCALED (BALANCED)



StandardScaler: Mean = 0, Variance = 1.

STRUCTURAL INTEGRITY: THE SPLIT



TRAINING SET (Pattern Recognition)

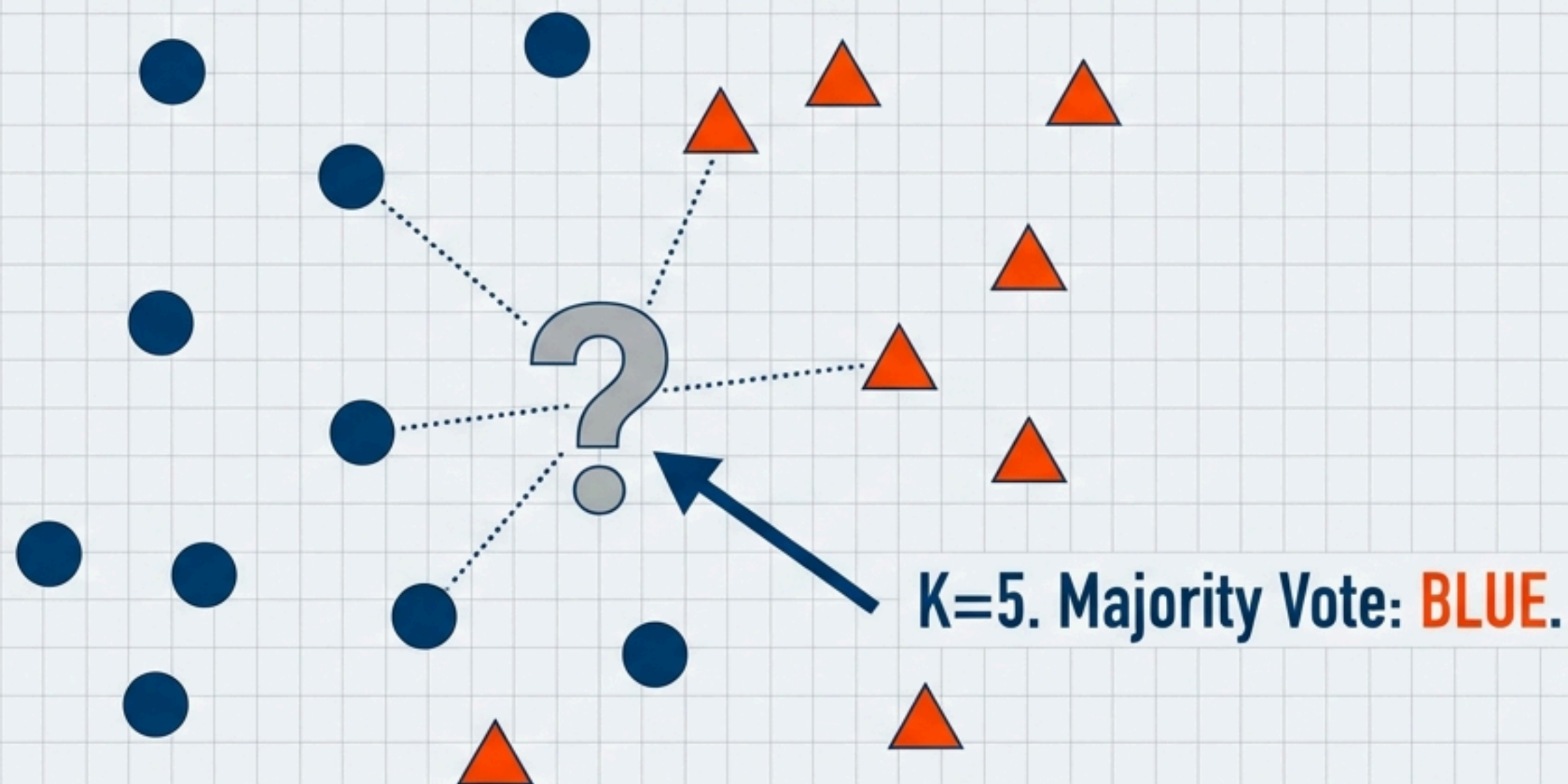
TEST SET
(Validation)



Randomize before splitting to remove order bias.

THE ALGORITHM: K-NEAREST NEIGHBORS

The Proximity Principle: Similar things exist in close proximity.



TUNING THE ENGINE: CHOOSING "K"



THE WORKFLOW: SCIKIT-LEARN

× ○ ○

```
model = KNeighborsClassifier(n_neighbors=5)
```

```
model.fit(X_train, y_train)
```

```
predictions = model.predict(X_test)
```

— **INSTANTIATE** (Build the frame)

— **FIT** (Memorize the map)

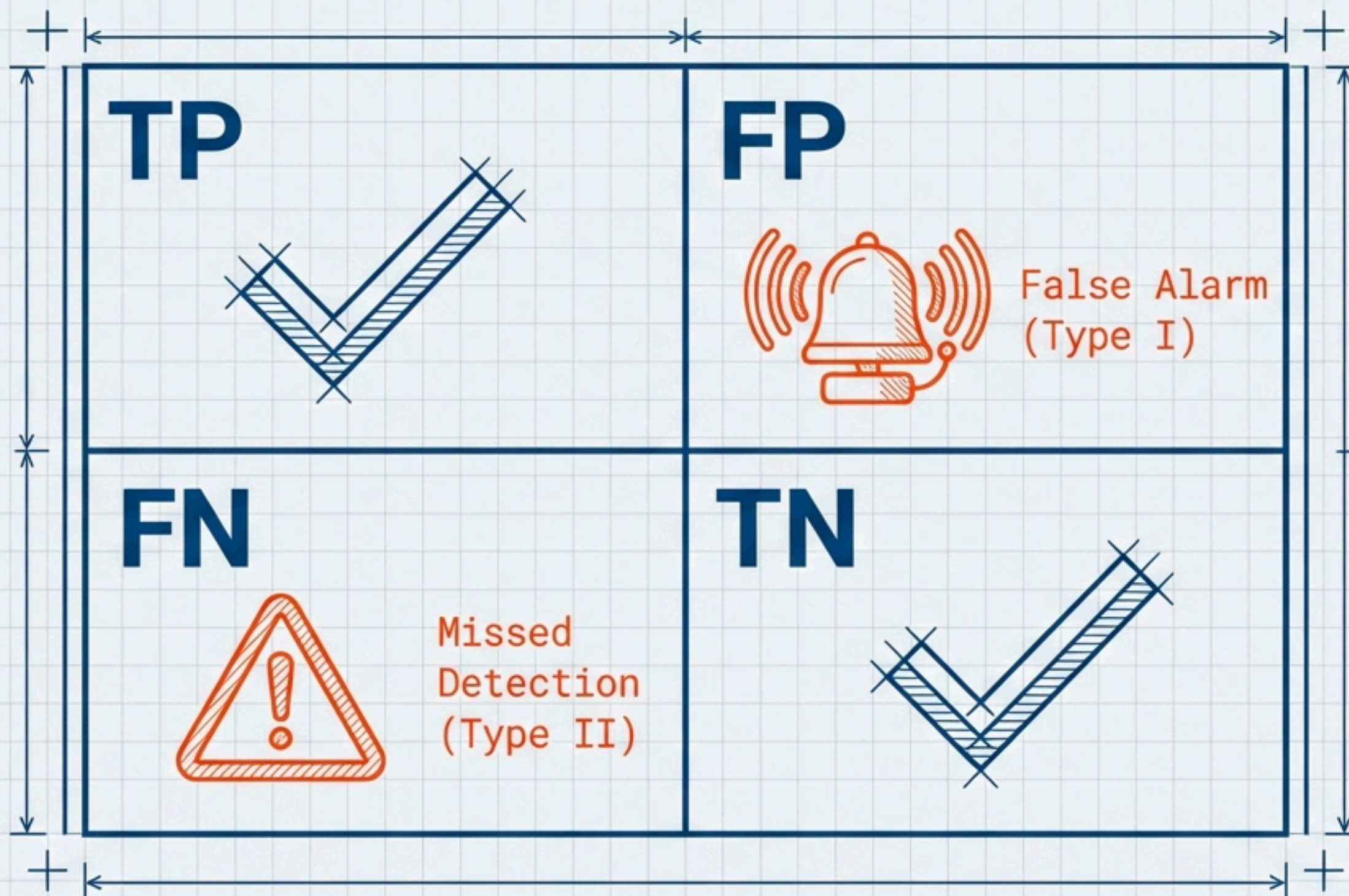
— **PREDICT** (Apply logic)

OUTPUT VALIDATION

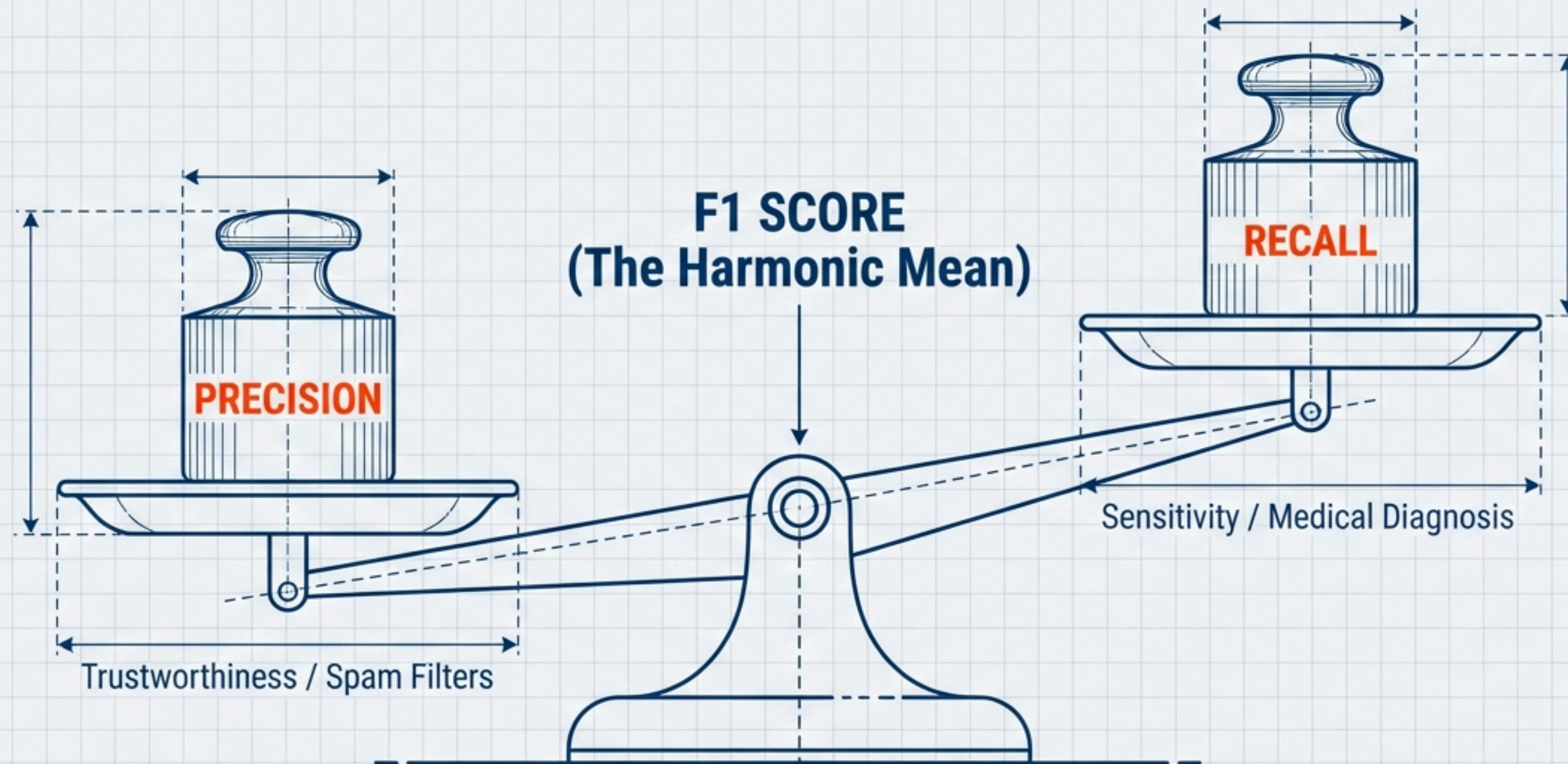


In imbalanced data, accuracy is a lie.
We must look deeper.

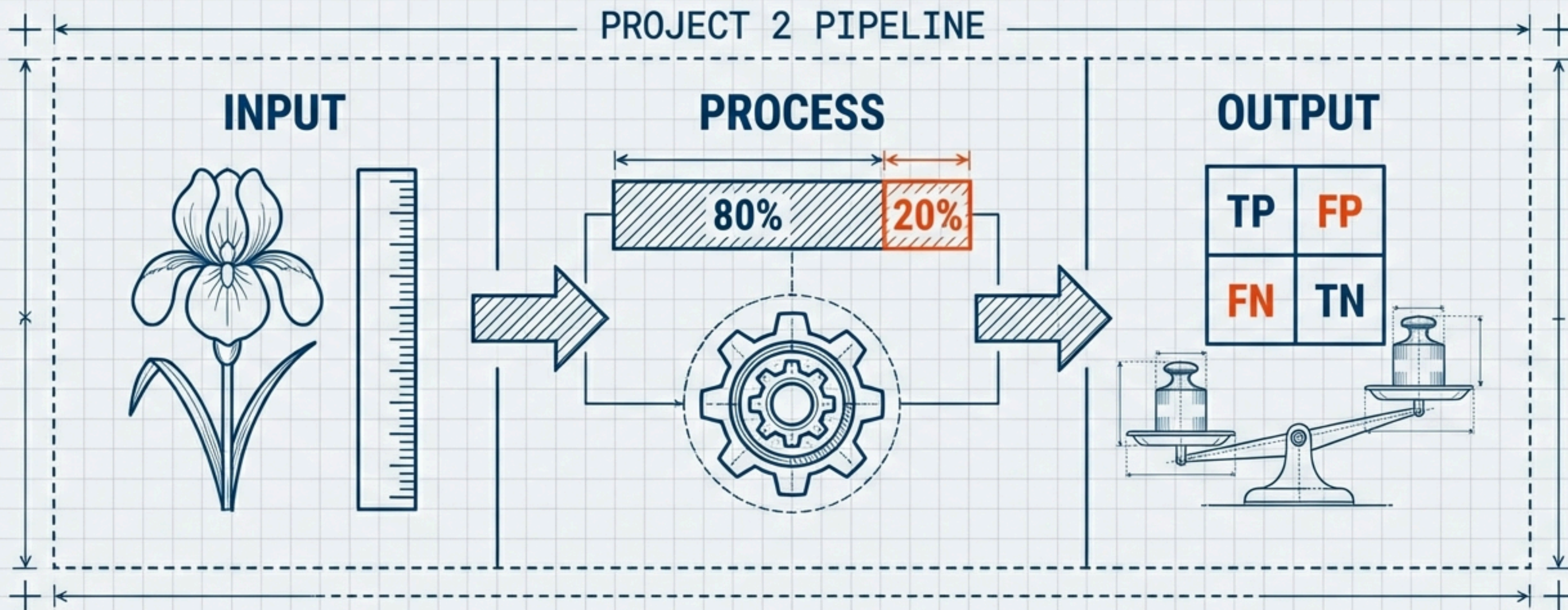
THE DIAGNOSTIC TOOL: CONFUSION MATRIX



STRATEGIC TRADE-OFFS



THE FULL ARCHITECTURE



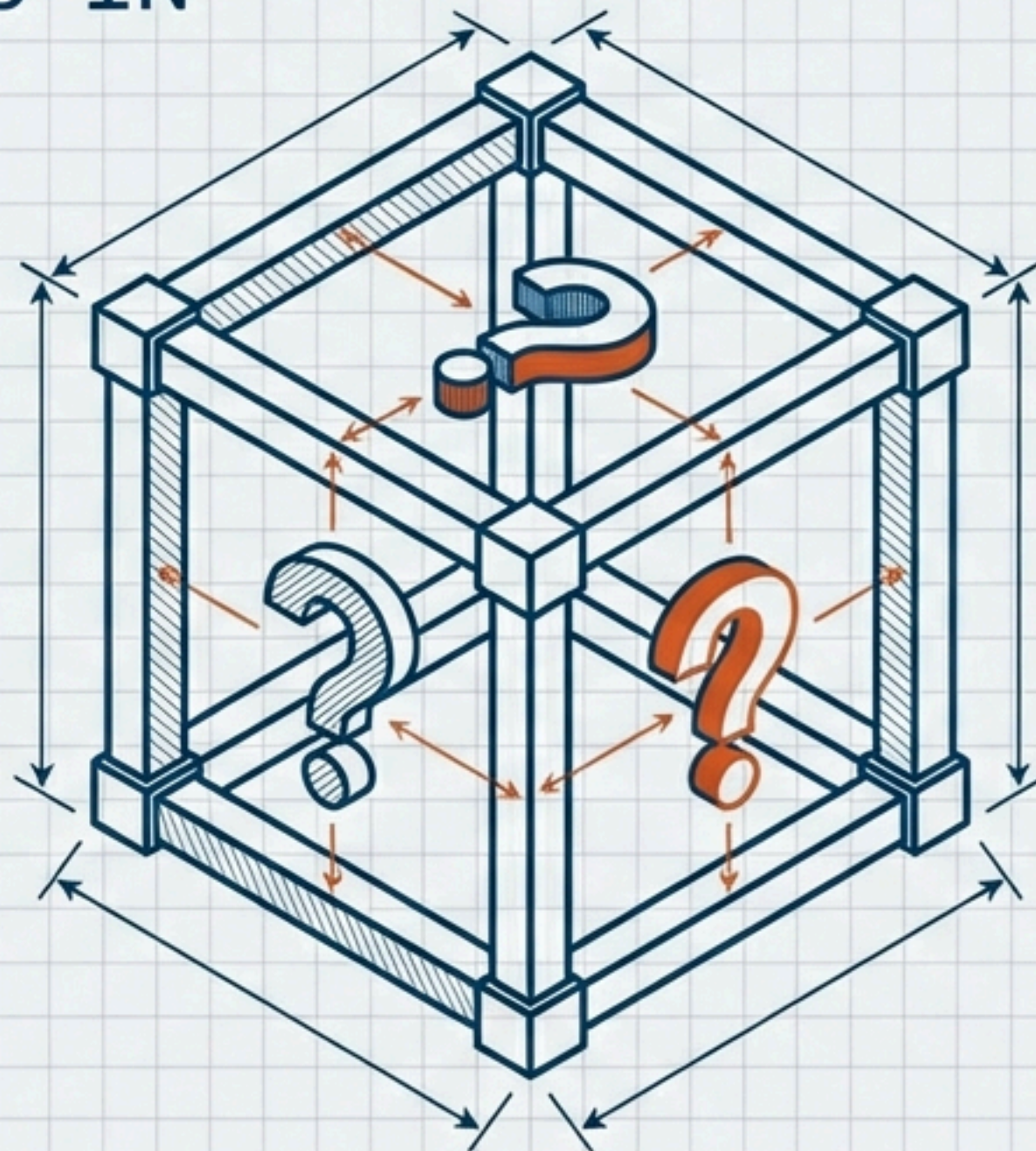
EMERGING HORIZONS



Next: Deep Learning & CNNs

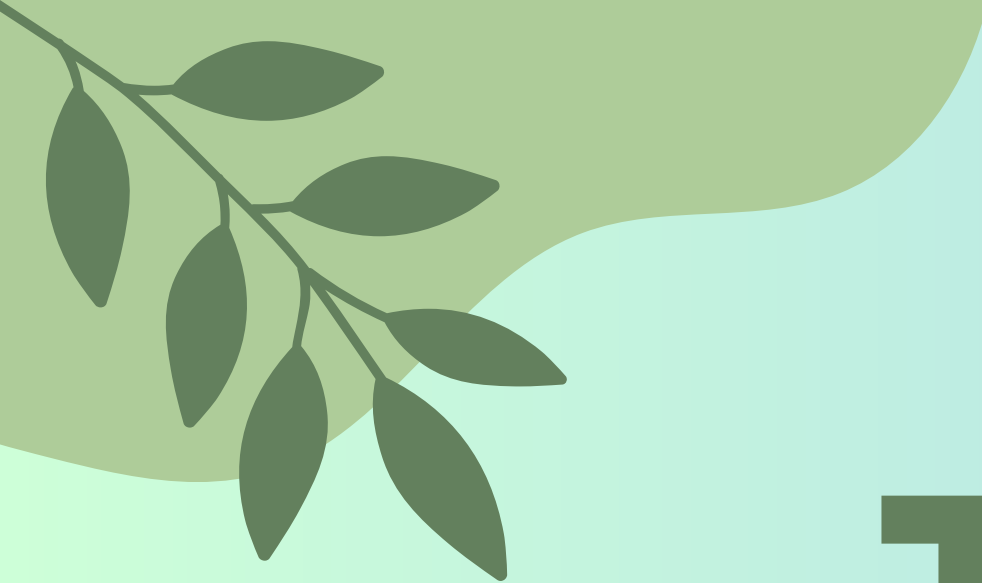
Q & A

THE ARCHITECT IS IN



CONCLUSION

The absolute best way to master Artificial Intelligence is through hands-on practice, not just theory. Don't just aim to complete these projects; take them one by one, experiment with unique solutions—like comparing different algorithms or testing your model with completely new data—and treat every low accuracy score as a learning opportunity. As you build these milestones at DecodeLabs, you are creating a real-world portfolio that showcases your supervised learning skills to future employers. Your journey to becoming a professional AI engineer begins right here, right now, with the very first model you train today.



THANK YOU

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